

Perspective

Foundation models for the electric power grid

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<https://doi.org/10.1016/j.joule.2024.11.002>

CONTEXT & SCALE Foundation models (FMs), pre-trained on large datasets and adapted to a broad set of applications, are revolutionizing the field of artificial intelligence (AI). Powerful FMs for language and weather have recently emerged, proving that such models can be developed for complex systems. Here, we conceptualize FMs for the electric power grid to be trained on grid data. It is shown that grid FMs have the potential to cope with the increasing complexity and uncertainties stemming from the energy transition by providing a significant speed-up in computation of at least 3–4 orders of magnitude. A key benefit is the generalizability of FMs that enables stakeholders to readily fine-tune the model for specific needs and their own proprietary data in a scalable and economical way. These capabilities make the FM approach ideal for unifying data, technology, and industry expertise toward a common goal. To develop such FM capabilities, we envision the power system and AI communities joining forces in a “moonshot” to develop transformative AI capabilities for the power grid, thereby supporting the energy transition.

SUMMARY

Foundation models (FMs) currently dominate news headlines. They employ advanced deep learning architectures to extract structural information autonomously from vast datasets through self-supervision. The resulting rich representations of complex systems and dynamics can be applied to many downstream applications. Therefore, advances in FMs can find uses in electric power grids, challenged by the energy transition and climate change. This paper calls for the development of FMs for electric grids. We highlight their strengths and weaknesses amidst the challenges of a changing grid. It is argued that FMs learning from diverse grid data and topologies, which we call grid foundation models (GridFMs), could unlock transformative capabilities, pioneering a new approach to leveraging AI to redefine how we manage complexity and



uncertainty in the electric grid. Finally, we discuss a practical implementation pathway and road map of a GridFM-v0, a first GridFM for power flow applications based on graph neural networks, and explore how various downstream use cases will benefit from this model and future GridFMs.

INTRODUCTION

Power grids are arguably the largest machines ever built by humanity, as recognized by the National Academy of Engineering.¹ Having survived the deregulation revolution at the dawn of the new century, this aging and ever-expanding system is about to face its biggest challenge. The energy transition, with its widespread electrification and massive deployment of renewable and distributed generation, is the main driver for this change, affecting both the demand and supply sides of the market.^{2–4} Furthermore, it is happening quickly. Climate change, manifesting with more frequent high-intensity weather events, poses additional challenges to system planners and operators.^{5–7}

In this context, it is clear that the operation, control, and planning of power systems will soon be pushed to their limits. As depicted in Figure 1, new computational methods and approaches are needed to better tackle the challenges posed by increased uncertainty and complexity. Machine learning (ML) and artificial intelligence (AI) methods have shown promise for such purposes in a wide spectrum of industries, with significant breakthroughs in computer vision,⁸ natural language processing,⁹ and intelligent control.¹⁰ Such approaches rely on statistical algorithms to learn from massive data, often enabling them to execute tasks (e.g., regression, classification, detection, segmentation, forecasting, and generation) without human supervision.

Yet, despite significant progress in the application of AI/ML methods to power systems,^{11–13} wider adoption has been impeded, particularly by a lack of readily available training data and the limited transferability of developed models to adjacent applications and system configurations.¹⁴ This is where, we argue, foundation models (FMs) have an important role to play. FMs are advanced AI models developed through self-supervised learning, most often based on transformer architectures, that generalize across various tasks after initial training on large datasets, enabling efficient adaptation to specific applications with minimal labeled data. Text-based FMs like GPT-4,¹⁵ encoding deep relationships in text, have enabled a wide range of new applications. Similarly, we anticipate that FMs for power grids, trained on the diverse data that characterizes power grids, will enable a broad set of new use cases.

In this article, we advocate for developing FMs for power grids: what we refer to as GridFMs. We explain why we believe in their potential, highlighting their anticipated strengths and weaknesses in light of the challenges posed by the evolving power grid. We introduce a practical road map for implementing GridFM-v0, an initial version of a GridFM for power flow applications, and discuss our vision for future GridFMs, pioneering a new approach in leveraging AI to manage complexity and uncertainty in the power grid.

In the following, we first highlight, in the “[looming challenges](#)” section, the looming power grid challenges associated with the energy transition, which are further aggravated by aging infrastructure, swelling cybersecurity, and climate change

concerns. We show how, by increasing the complexity and uncertainty of the power grid, these drivers are creating a significant gap in computational capabilities, which we propose to overcome with FMs. We then review in the “[foundation models](#)” section the strengths and weaknesses of FMs and propose in the “[proposed implementations for GridFM](#)” section the near-term and long-term development phases of GridFMs that we argue will overcome current limitations and enable scenarios, which are currently unattainable.

LOOMING CHALLENGES

As the energy transition accelerates, power grids face unprecedented challenges threatening their reliability and efficiency.

Distributed and renewable energy sources

Although a single coal plant can generate hundreds of megawatts predictably and reliably, large numbers of weather-dependent renewable energy resources, such as wind and solar, are required to produce the same power output. Current methods to forecast 1 day or 1 h ahead still show forecast errors of up to 10% and 3%, respectively,¹⁶ depending on the energy source. Additionally, distributed energy resources (DERs), such as battery storage and electric vehicles, are operated by independent power management systems and are driven by user patterns with intrinsic uncertainties. Dealing with the added uncertainty from reverse DER power flows, multipoint injections, and weather-induced fluctuations requires new voltage regulation methods, protection schemes, and advanced (optimal) power flow analysis. The numbers of controllable and dispatchable devices, as well as balancing nodes, are expected to grow exponentially,¹⁷ with additional thousands to millions to be added in the near future. Managing the associated variability and uncertainty will require advanced stochastic optimization, scenario reduction, and sampling-based approaches, which are computationally more intensive than currently used deterministic methods, especially for real-time demand-supply balancing. Furthermore, as the net load of DER-rich distribution branches seen by transmission operators becomes more variable and less predictable, effective grid management and planning will become impossible without effective co-simulation frameworks that integrate transmission and distribution analysis. Yet, the computational cost of such simulations, at least with existing technologies, is extremely high.

Everything inverter

For the future power grid, inverters will play a crucial role in maintaining low cost at the grid's edge. However, with increased inverter proliferation, established control schemes and grid stability are challenged, while a more deregulated environment is created. With more inverters connected, grid inertia—which stabilizes phase, voltage, and frequency during transient events and

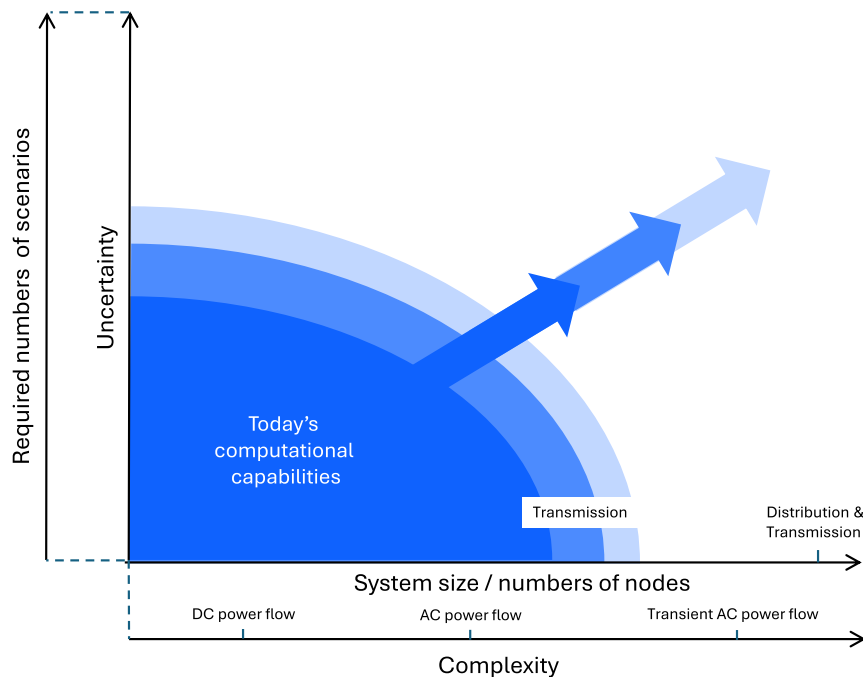


Figure 1. The energy transition, aging infrastructure, cybersecurity challenges, and climate change greatly increase complexity and uncertainty in operating, controlling, and planning the power grid

This is creating a widening gap between existing computational capabilities and the evolving needs of the electric power industry.

Changes to demand and weather patterns

Mass electrification of heating, transportation, and industry—in addition to climate change and shifting demographics—result in changing load profiles. Consequently, the accuracy of classical short-term load forecasting has declined over the past decade due, among other factors, to changes in customer demand and weather patterns, leading to out-of-distribution scenarios.²⁸ To model demand uncertainty, utilities use multiple scenarios for structural and random drivers.²⁹ Increasingly, they also forecast at the power transformer level to manage

is mostly tied to large synchronous generators—decreases,¹⁸ unless more grid-forming converters are introduced. Because inverters are controlled by software, they can quickly adjust power or disconnect, causing additional issues such as sub-synchronous resonances and even trip large generators.¹⁹

Despite the increasing adoption of standards,²⁰ inverters remain hard to predict and model, unlike physics-driven synchronous generators. Quasi-static phasor simulations are used to assess optimal power flow. For transient grid stability assessment, however, computationally expensive electro-magnetic transient (EMT) simulations²¹ are required. They run mostly on traditional central processing units (CPUs), which are flexible for many algorithms but inferior in computational performance to recent graphics processing units (GPUs), which are just starting to get adopted in the industry, as, e.g., in Sun et al.²² Furthermore, CPU systems do not always scale well for larger EMT simulations when run on multiple CPU cores.²³ For example, consider ParaEMT – NREL’s EMT simulator^{23,24}; on a single CPU, it runs a 1-s, 50- μ s resolution EMT simulation for the WECC 240-bus system in 29 s. Another speed-optimized algorithm²¹ takes 11 min to compute a 0.25-s window for a 52-bus system at 1- μ s resolution. Grid operations, however, would require EMT simulations to run in real-time and for large sections of the grid, including distribution grid sections.

To further account for inherent network uncertainties and those introduced by inverter-based resources (IBRs) and DERs, stochastic ensembles of the problems must be simulated. Altogether, such an approach is computationally very expensive and, with current approaches, impractical.²⁵ Thus, the community mostly relies on offline simulators, such as PS CAD²⁶ or ParaEMT,²³ while starting exploring Physics-informed neural networks (PINNs) to reduce EMT simulation time.²⁷

local dynamics,³⁰ but this approach is costly and does not scale due to the need for multiple models.

Short-term models

Short-term models for operations, running hourly to weekly, are used for market bidding, unit commitment/economic dispatch, operational planning, and security analysis, among others.³¹ As stated before, uncertainty and variability are driven in part by weather and human behavior, from societal aggregations to individual households,³² as well as by operator-induced industrial activities. Economic factors, the increasing frequency and intensity of extreme weather events with high prediction uncertainty,³³ and growing system complexity (e.g., self-consumption and electric vehicles) also contribute to variability. The resulting complexity necessitates the selection and integration of new and heterogeneous data sources (e.g., social networks and online news) along with the frequent adjustment, rebuilding, and maintenance of models.

Long-term models

Long-term models, such as the regional energy deployment system,³⁴ are used for resource, infrastructure, and strategic planning. Uncertainty and variability at longer time scales are significantly driven by climate policy, technology adoption, and regulatory changes. Instances of this include the combustion engine phase out by 2035 in Europe—which resulted in a massive increase in the long-term electricity demand forecast—or the US Federal Energy Regulatory Commission’s rules on planning and covering the cost of the power grid.³⁵ To improve model accuracy, additional information that captures demographic changes and changes in the distribution of wealth, which lead to different societal and individual behaviors (e.g., heating, cooling, and electromobility), as well as climate-change-induced environmental shifts, must be integrated. Therefore, the capacity to

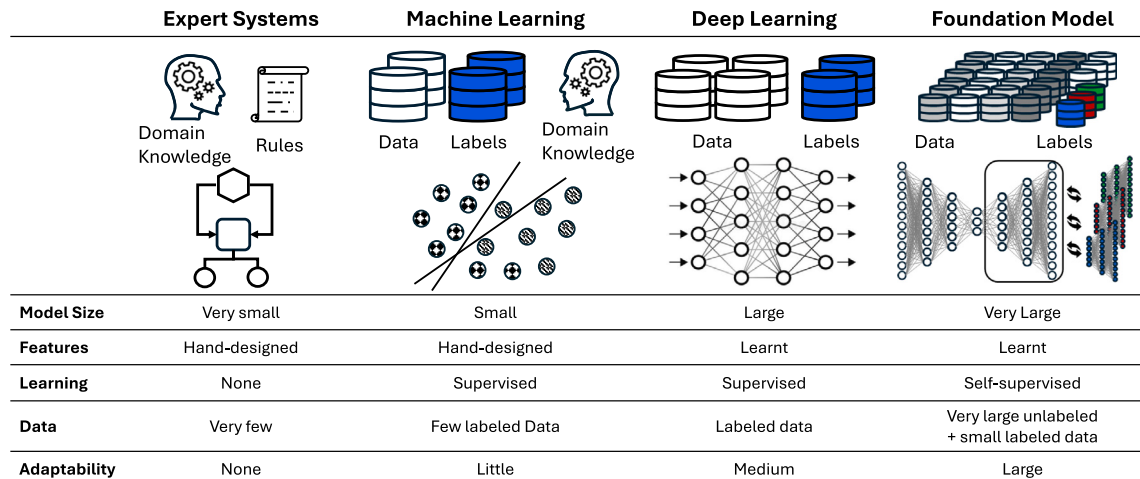


Figure 2. Evolution in AI and ML methods, leading to the emergence of FMs

combine heterogeneous data sources, including information in textual format, is a fundamental requirement to improve predictability.

Aging infrastructure

Managing the vast amount of diverse equipment and aging infrastructure—the US power grid has hundreds of thousands of miles of transmission and millions of miles of distribution lines with over 55,000 substations—constitutes a major challenge on its own. Although some lines last up to 80–100 years,³⁶ insulators, transformers, and generators require frequent inspections, ideally through permanent sensing, which, given their age and approaching end of life,³⁷ may not always be installed. Modeling these components in mostly unknown conditions is difficult and may rely on computationally expensive stochastic and ensemble models. Driven by growing demand, utilities are increasingly exploiting their aging infrastructure toward its operational limits,³⁸ reducing margins and leading to larger size contingency analysis problems.³⁹ In response, much faster grid analysis functions are required, e.g., including grid contingency events.

Moreover, the lowering of grid inertia through IBR integration (see “everything inverter” section) is further amplified by the retirement of old synchronous generators. The more frequent use of synchronous generators for regulating purposes brings them more often into operating zones where mechanical or thermal stresses are significant, thus accelerating the aging of the generators. IBR-related switching transients also disrupt existing protection relay schemes⁴⁰ and increase grid behavioral uncertainty.

Cybersecurity threats

DER and IBR technologies are currently being developed and deployed without robust regulatory frameworks mandating essential cybersecurity standards, hence increasing the associated cybersecurity threats. Interconnection and interoperability standards, such as IEEE Std. 1547, requires DERs and IBRs to communicate with clients, vendors, aggregators, and grid oper-

ators for monitoring and control. In practice, however, communication often occurs over inadequately secured public and private networks, expanding the cyberattack surface.⁴¹ Consequently, critical vulnerabilities are frequently discovered in commercial DERs and IBRs, such as electric vehicle charging stations⁴² and PV inverters,⁴³ increasing the likelihood of major cyberattacks.⁴⁴ As the penetration of DERs and IBRs increases, synchronized cyberattacks targeting DERs or IBRs could cause grid reliability issues, from localized instability to system collapse.⁴¹ This risk is exacerbated by reduced grid inertia and contingency reserves due to reliance on zero marginal cost renewables. Additionally, control and management responsibilities for DERs and IBRs are often delegated to aggregators, vendors, and customers. This delegation complicates risk assessment, incident detection, and response—as visibility barriers are created for grid operators—introducing uncertainty and black-box behavior for large sections of their grid. With the lack of real-time data, these barriers create significant additional computational burdens for automated and accelerated data analysis to search for fraudulent injections.

FOUNDATION MODELS

Having identified the growing gaps between computational capabilities and industry needs amidst the energy transition, we now introduce FMs, laying the groundwork for our vision of GridFMs.

The FM approach

To understand the FM approach, it is helpful to review the evolution of AI and ML as depicted in Figure 2. Early hand-crafted symbolic expert systems, from the 1960s onward, were limited and brittle. The 90s brought big data and general-purpose ML that, however, still often depend on task-specific hand-crafted features. In 2012, deep learning, enabled by increased computational performance, started to disrupt the AI world⁴⁵ but was more often than not used to train models on large, annotated datasets in a supervised way, making them task-specific.

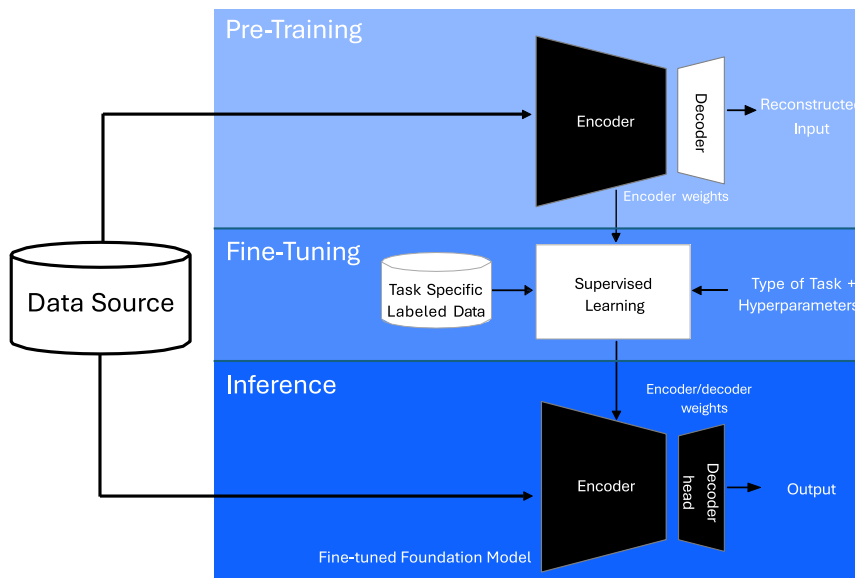


Figure 3. FM life-cycle phases (1) pre-training, (2) fine-tuning, and (3) inference

FM pre-training encodes underlying patterns from the supplied data in model parameters. Successful examples of FMs to date include text-based large language models (LLMs)⁴⁷ and also weather forecast models,⁵⁰ in which these patterns are linguistic and Physics-driven temporal-spatial relationships, respectively. LLMs have been shown to be applied successfully to certain power system problems.¹⁴ By contrast, our focus here is on GridFMs that go beyond text, which will eventually be able to handle multiple data modalities intrinsic to power grid operations. Modalities of interest include network topologies, sensor data from phasor measurement units (PMUs),

alarms, SCADA systems, advanced metering infrastructure (AMI), load, generation, weather, and market data.

Recently, FMs have emerged. They learn from data through self-supervision and large content-rich datasets,⁴⁶ and have proven to generalize across many applications. FMs draw their potential from being able to train on very large datasets, which have not been annotated and thus are more easily available than labeled datasets, as required for supervised learning. The FM approach is a three-step process, as shown in Figure 3:

Pre-training

A general-purpose FM is developed through self-supervised pre-training and a suitable architecture, often based on a transformer model with encoder and decoder components. This step involves a reconstruction task where the model reconstructs masked parts of the data.

Fine-tuning

The pre-trained FM is customized for various downstream tasks and applications with minimal labeled data by attaching a task-specific decoder head and performing a few training iterations.

Inference

The fine-tuned FM is deployed to allow users to request predictions at low computational cost.

The development of FMs involves extensive pre-training on datasets of tens of terabytes and models with hundreds of millions or even billions of parameters.⁴⁷ However, in contrast to what one might assume, even bespoke AI models can be very complex and expensive to train. For example, the non-foundation and application-specific weather AI models, Graphcast (for forecasting) and CorrDiff (for downscaling), were trained on 32 TPU devices⁴⁸ and on 128 A100 GPUs,⁴⁹ respectively.

The key advantage of FMs is that once developed, they can be readily adapted to multiple downstream applications via fine-tuning with minimal labeled data. Thus, the cost of model pre-training can be spread across numerous applications. These economies of scale make it feasible to employ AI for niche applications, for which developing a dedicated traditional AI model might be prohibitively expensive or technically infeasible.

Salient advantages

We next review salient features of AI in general and FMs in particular that hold significant potential for enhancing power grid applications.

Predicting “tokens”

In transformer models, a token is defined as the smallest fundamental unit of data (e.g., a character sequence or image patch). FMs have been shown to excel at predicting missing tokens based on context, enabling tasks such as sentence completion, programming assistance, and question answering in the text domain.⁵¹ This capability extends beyond text to fields such as time-series, audio, or video, where FMs predict the next or missing values.⁵²

Homogenization and scaling

FMs represent a paradigm shift in AI, offering a scalable and standardized approach and making niche applications commercially viable. This AI-as-a-platform approach promises to simplify operations by allowing a few FMs to replace many specialized models. For example, a single geospatial FM for earth observation has been fine-tuned for tasks such as flood segmentation, land-use classification, and heat island regression.⁵³

Accelerating simulations

AI models, in general, including FMs, can emulate physics-based simulations with impressive accuracy and efficiency, potentially combined with physics-informed ML.²⁷ The efficiency gains originate from AI models directly mapping inputs to outputs, unlike traditional physics-based simulations that solve partial differential equations iteratively. Recent AI models⁵⁴ emulate numerical weather prediction models⁵⁵ with comparable forecasting skills but up to 4–5 orders of magnitude lower computational cost⁵⁶ during inference. These game-changing advances improve the spatial and temporal resolution of forecasts and, combined with AI techniques such as diffusion-based models,⁵⁷

facilitate the creation of large ensembles for uncertainty estimation. Another salient advantage of using AI for emulating simulations lies in the fact that no *a-priori* assumptions are required, but rather the AI model can learn the full physics directly from the data or observations. We argue that exactly these characteristics of AI in general and FMs in particular are crucial for bridging the computational gaps shown in Figure 1.

Hurdles to implementation

Here, we discuss potential hurdles associated with implementing FMs for the power grid while suggesting solutions.

Need for data

FMs, similar to other deep learning models, are data hungry. For example, the Llama3 LLM, with 70 billion parameters, is trained on 15 trillion tokens of text.⁵⁸ By way of comparison, the power sector globally generates several times that volume daily.⁵⁹ Considering the groundbreaking capabilities of LLMs in open-ended systems where context and semantics are not consistently described by mathematical equations, we are confident about this technology's potential for the power grid. Diverse and high-quality datasets required for training grid FMs will, however, still have to be assembled to cover the full spectrum of scenarios and situations. Addressing transmission grid problems first, where data are available in a more standardized form, e.g., PMU or power flow data, will help demonstrate the value of FMs for the grid. This will create incentives for data at the distribution level, which currently are much more heterogeneous and of less quality,⁶⁰ to be collected, curated, and made available in suitable formats for use with AI and FMs at a larger scale.

Data accessibility

Data accessibility is another hurdle to overcome. Grid data are heterogeneous and distributed, are owned by multiple entities, and have special security and privacy measures. A potential solution to this barrier includes privacy-preserving federated learning⁶¹ and secure enclave technologies.⁶² Such approaches may enable multiple data owners to train models collaboratively without sharing their local data with a central server, thus preserving privacy and complying with data protection regulations. Additional schemes, such as differential privacy and homomorphic encryption, can be integrated into federated learning frameworks,⁶³ ensuring that sensitive information remains protected during training. Secure enclaves, in contrast, create a trusted execution environment for secure processing of sensitive and proprietary data during FM training. Moreover, simulated data from real and synthetic systems can complement and provide scenarios under various conditions, such as extreme weather conditions, component failures, and cascading failure events.¹³ As discussed further in "propose implementations for GridFM" section, GridFMs might be developed by synthetic data, which are then updated and verified with proprietary data behind the user's firewall. However, it is important to consider that electrical networks evolve for various reasons, such as network reinforcement, which changes topologies and operating conditions, or new business models (e.g., local energy communities), which change the operating profiles of assets connected to the network, requiring continual learning of the FM.

Trust

Notwithstanding the scenario in which all necessary data, whether synthetic or real, can be cleared and collected for use, there still remains the age-old question of whether the data and, consequently, a data-driven model can be trusted. In general, power system operators prefer interpretable models over black-box models so that the root causes of choices by advanced tools can be better understood, and heuristic solutions can be developed. Although the debate on the trustworthiness of AI will continue, there is evidence that AI models can match the performance of first-principle physics-based models, at least for specific applications.⁶⁴ Performance alone, however, does not seem to be necessarily a sufficient condition for trust because even conventional solvers can have stability and convergence challenges,⁶⁵ which we have found ways to deal with through decades of usage.

Referring back to the recent breakthroughs of AI emulators for weather, it was demonstrated that AI could, indeed, encode the actual dynamics of certain physical systems and generalize to unseen events in ways that allow it to have comparable performances to a physics- and simulation-based model.⁶⁶ In some cases, e.g., for an AI power flow emulator, which is discussed in "propose implementations for GridFM" section, the AI results can be immediately verified by using the physics equations. In other cases, where such ad-hoc verification is not possible, a consistent way is required to evaluate the model for out-of-sample scenarios. In the longer term, a set of standards must be developed as to what performances must be achieved for a given out-of-sample scenario. Toward that end, ISO has already developed the first family of standards, e.g., ISO/IEC 24029-2.⁶⁷ The European Union's recently published AI Act⁶⁸ defines two concepts to boost trust in AI-based systems and data sharing: (1) testing and experimentation facilities that combine physical and virtual environments to evaluate and certify their latest AI-based software and hardware and (2) the data spaces concept aims to remove barriers to data sharing, such as interoperability, trust, and privacy, while keeping sovereignty and full control of data.

Malicious use

FMs can be used as a sandbox to accelerate the development of adversarial attacks. For example, FMs' potential for scalable data imputation, combined with their ability to predict grid operating conditions, could be exploited by adversaries to infer sensitive information about a target power system.⁶⁹ Therefore, it is imperative to continuously assess the extent to which misusing such technology can augment malicious actors' attack capabilities and to adapt cybersecurity defenses and policies in response.⁷⁰

PROPOSED IMPLEMENTATIONS FOR GridFM

The broader vision of a GridFM is the enablement of all or as many applications as possible. However, even if initially GridFMs would only be suitable for a single application, the superior adaptability of FMs over bespoke AI models through fine-tuning to thousands of different utilities with their specific data and requirements justifies the FM approach.

APPLICATIONS			
Transient and dynamic stability • Time-domain power, frequency • Rotor angle stability • Fault currents	Load flow • Power flow • State estimation • Optimal power flow	Forecasting • Demand • Renewable generation • Electricity price	Control operations • Frequency & voltage control • Network reconfiguration • Protection activation
Electricity markets • Market dispatch • Economic & Re-dispatch • Unit commitment	System security • Fault detection • N-1, N-K analysis • Cascading failure analysis	Expansion planning • Scenario generation • Grid upgrade • Generation expansion	Cyber security • Intrusion detection • Vulnerability assessment • Incident Response Plan

Figure 4. Applications: examples of power system problems to be solved with GridFMs

In the above context, fine-tuning can mean different things. A GridFM may be fine-tuned to a set of diverse applications, such as load forecasting, fault detection, and others. Initially, this may be too ambitious and challenging, considering the diversity of downstream applications as shown in Figure 4, each requiring different data, temporal scales, and pre-training strategies. But fine-tuning may also refer to tailoring a GridFM for a given application with user-specific (and potentially private) data to the particular user needs and requirements. For example, just in the United States, there are 3,200 utilities,⁷¹ with their own private and siloed data and grid-specific requirements. These utilities may not have sufficient data by themselves but could directly profit from the proposed approach. This multi-level fine-tuning approach is not unlike what can be observed in geospatial or weather FMs, where FMs have shown excellent adaptability to different geographic regions and spatial and temporal resolutions for a given downstream application.^{50,72,73}

We thus propose an implementation road map as outlined in Figure 5, with a first, more specific version of GridFMs, pre-trained for power flow applications (GridFM-v0), which is expected to provide real-world benefits in the next 1–3 years. In the long term, we envision expanding and addressing more grid applications by including more diverse data modalities to pre-train a single GridFM or by combining different GridFMs, as discussed in the “long-term vision: GridFMs for a broader set of applications in power systems” section. The rationale to start with power flow is as follows:

A highly relevant pre-training task

FMs are known to excel when the pre-training task is closely related to the enabled downstream applications.⁷⁴ Consequently, with AC power flow at the core of power system analysis,⁷⁵ its reconstruction is an ideal pre-training task for GridFM-v0. This enables power-flow-related applications in the area of load flow (optimal power flow, state estimation, and power flow), system security (contingency analysis and cascading failure), and beyond.

AI power flow models are computationally more efficient

Current power flow solvers, e.g., using Newton-Raphson methods, are computationally expensive, especially for large grids. For instance, for an N-1 contingency analysis on the Western Interconnection, it takes approximately 5 min per scenario,⁷⁶ whereas meeting the NERC TOP-001-3 requires running these analyses within a limited time period. In contrast, GridFM-v0 is

projected to achieve a computational speed-up of 3 to 4 orders of magnitude over conventional solvers, as demonstrated on IEEE grid cases with up to 118 buses.⁷⁷ For larger grids, we anticipate a similar or greater speed-up because GNNs exhibit linear complexity in the number of grid buses,⁷⁸ whereas Newton-Raphson methods scale quadratically with the number of buses.⁷⁹ This significant efficiency gain will enable real-time analysis of multiple scenarios of more complex grids,⁷⁷ which will become crucial for maintaining system stability and optimizing future grids under uncertainties stemming from the energy transition and from climate change.⁷⁶

There have been significant advances in GNN-based power flow

Using GNNs to represent the power grid and mapping power flow to the message-passing scheme of GNNs has been successfully demonstrated recently.⁷⁷ These models replicated the results of traditional power flow solver with less than 1% error on the predicted quantities despite past GNN training challenges.⁸⁰ Although these models were trained in a supervised way, recent results of the previously discussed GNN-based weather FMs^{50,72,73} indicate that learning power flow with GNNs in a self-supervised way is a valid approach for a GridFM. Furthermore, large-scale data FMs based on GNNs have already been developed,⁴⁸ and a comprehensive review can be found in Mao et al.⁸¹

Near term: GridFM-v0, a GridFM for power flow applications

The four phases as foreseen for the development of GridFM-v0, and as depicted in Figure 5, are discussed in detail in the following sections.

Phase 1: Identify, collect, and generate training data

Training GridFM-v0 requires a large collection of solved power flow problems for different grid topologies, parameters, and different load conditions. In recent years, large power grid datasets have been made available, created by (1) collecting grid topology and measurement data, (2) generating varying synthetic load conditions per grid model, and (3) using established solvers to compute a wide range of power flow solutions under different operational conditions. We aim at using the dataset provided by Varbella et al.,¹³ which contains power flow solutions under real load conditions on four different grid topologies. Additionally, to complement these data, diverse power flow problems using traditional power flow solvers^{82–84} will be

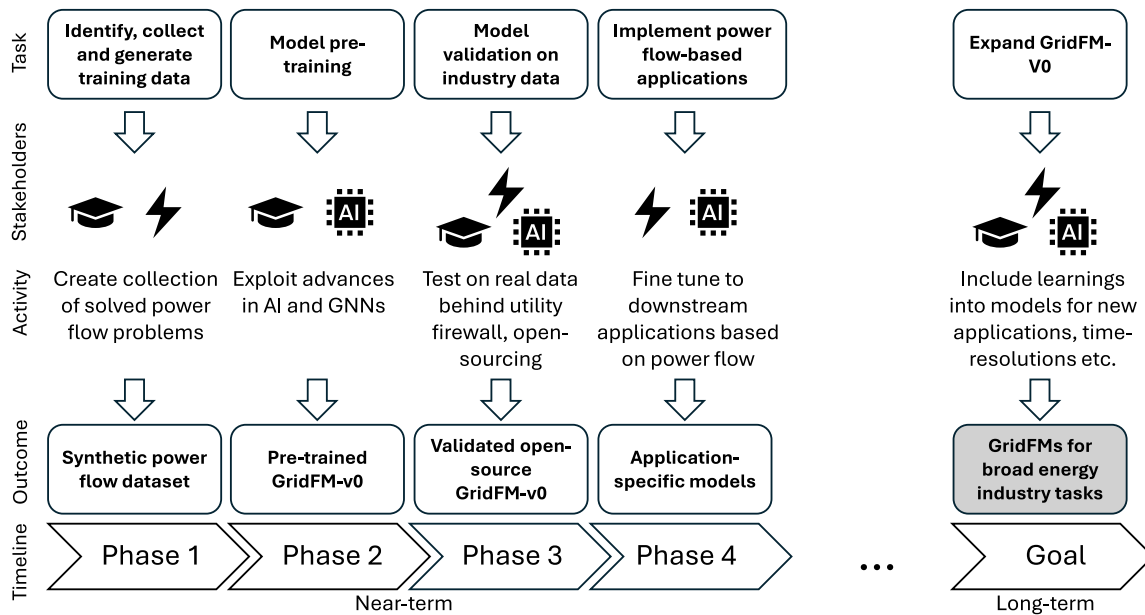


Figure 5. GridFM implementation road map with four near-term phases to develop GridFM-v0, a GridFM for power-flow-related applications
The icons refer to academia, power system industry, and AI community. The long-term goal is creating a family of GridFMs for a broader range of grid challenges.

computed, with power grid topologies from PGLIB-OPF⁸⁵ and standard IEEE benchmarks and with topology and load perturbations. Consequently, our model will be pre-trained on data from realistic topologies with a broad range of operational conditions. This approach facilitates the model to generalize well across topologies, as demonstrated in Varbella et al.⁸⁶ For model fine-tuning, datasets from Varbella et al.⁸⁶ for system security analyses and Lovett et al.⁸⁷ for optimal power flow (OPF) will be used. The latter contains 300,000 solved optimal power flow problems for 10 different grid topologies each, including solutions for N-1 contingency analysis, with grid sizes ranging from 14 to 14,000 buses.

Phase 2: Model architecture development

In the second phase, suitable architectures will be evaluated by pre-training performance analysis and by adjusting models, training strategies, and loss functions. Figure 6 illustrates the masked auto-encoder (MAE) pre-training task of GridFM-v0. Intuitively, the grid is modeled as a graph, where nodes represent electrical buses with active power, reactive power, voltage magnitude, and voltage angle, $(p_i, q_i, v_i, \delta_i)$, as node variables. We opted for this graph representation of power systems, but others can also be considered.⁸⁸ Transmission lines and transformers are modeled as graph edges. An autoencoder^{89,90} is trained in a self-supervised manner to reconstruct masked node features from power flow solutions, each corresponding to a specific topology and load condition. This amounts to solving the power flow equations. During training, the loss function ensures meaningful reconstruction of node variables within the physical constraints. The total loss $\mathcal{L} = \mathcal{L}_{SCE} + \mathcal{L}_{powerflow}$ combines the scaled cosine error (SCE),⁸⁹ which minimizes the distance between the original and reconstructed graphs, and $\mathcal{L}_{powerflow}$, which ensures that the power flow equations are satisfied.

Recent studies show that an approach using physics-informed methods is the most effective for solving OPF. For example, Chatzos et al.⁹¹ demonstrated that a hybrid loss function provides high-fidelity approximations for large-scale OPF, whereas Nellikkath et al.⁹² confirmed that this approach best solves AC OPF by integrating data-driven learning with physical laws. FMs, as a concept, are designed to incorporate diverse methods, including GNNs and PINNs, into a unified framework, enabling them to generalize effectively and adapt to various tasks, such as solving OPF problems.

Phase 3: Model scaling and validation on industry data

Due to regulatory constraints, it is unlikely or at least challenging to gain full access to sensitive data from operational power grids.^{93–95} To ensure the model's accuracy and relevance in practical applications, it is therefore imperative that a GridFM be validated on operational data and real grid topologies. To address these challenges, an iterative approach is adopted. Our GridFM-v0 will initially be pre-trained and tested on openly available data. Subsequently, it will be re-trained and validated in collaboration with electricity utility partners behind the firewall in a secure compute environment. The results of these validations will guide us to generate more suitable synthetic training data, update the architecture, loss functions, etc., so that the improved GridFM yields satisfactory results in real-world conditions once fine-tuned. This approach ensures that the GridFM remains performant and aligned with industry needs without exposing nor pre-training the model on proprietary data.

Phase 4: Implement power flow-based use cases

GridFMs are anticipated to achieve a computational speed-up of 3 to 4 orders of magnitude over traditional power flow solvers, as stated previously, which will directly benefit various applications.

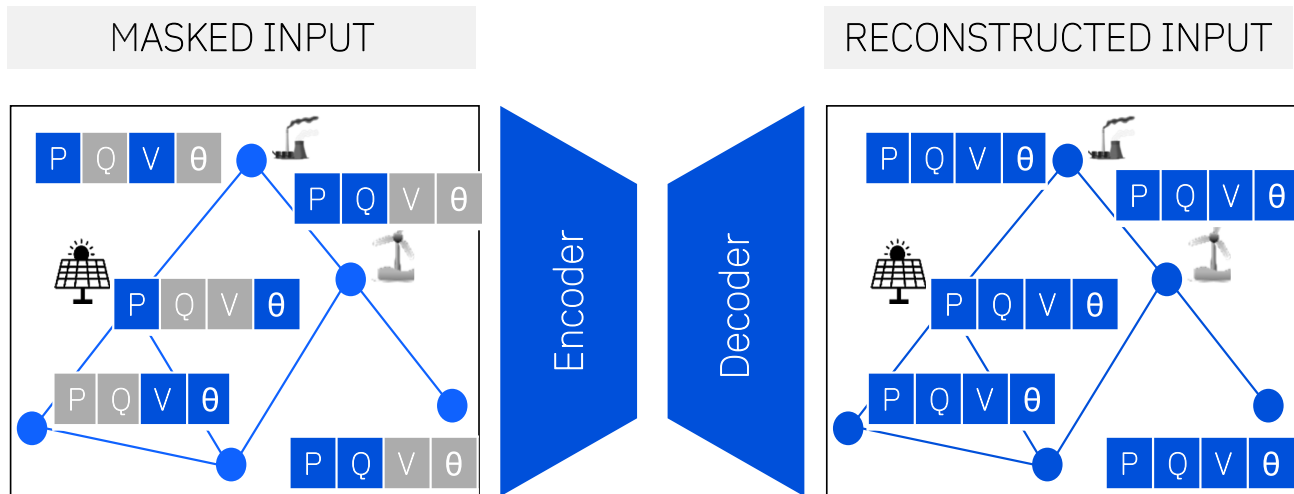


Figure 6. GridFM-v0 is pre-trained to reconstruct masked power flow data from input graphs

Complementing numerical power flow solvers. For applications that require solving power flow frequently, and where speed is critical, linear approximations (such as DC power flow⁹⁶) are typically used. For these applications, faster AI-based power flow solvers offer a trade-off between traditional DC power flow and AC power flow solutions. We expect minimal fine-tuning of these models because the task is equivalent to the suggested pre-training reconstructing task.

Contingency analysis. Contingency analysis is essential for assessing grid resilience. On a grid with 1,000 lines, simulating the loss of each individual line requires solving 1,000 distinct power flow problems. However, when considering scenarios that involve the simultaneous loss of any two components, the number of power flow analyses increases dramatically to $\binom{1,000}{2} = 499,500$. By utilizing a power flow solver that is 1,000 times faster, operators will be able to perform contingency analyses across a much larger set of scenarios. For example, solving 499,500 scenarios with AI-based methods would take roughly the same time as solving only about 500 scenarios using conventional physics-based solvers. Therefore, a GridFM-v0 will greatly enhance grid analyses and contribute to operating and planning under uncertainty.

The power grid is continuously evolving; therefore, the GridFM-v0 must be able to account for different network topologies and different operating conditions of a same grid. This can be accomplished by training the model on grids with topological perturbations, as already demonstrated by Piloto et al.,⁹⁷ or by additionally training the same model on entirely different grid topologies, as done in Varbella et al.⁸⁶

OPF. The authors in Piloto et al.⁹⁷ trained a model on synthetic data and demonstrated an OPF speed-up of 2 to 4 orders of magnitude while maintaining errors of less than 1% in the objective value. Solving the OPF translates into fine-tuning the GridFM for reconstructing the generator power to minimize generation costs. Reconstructed node variables must adhere to bounds (e.g., generator capacity constraints), making this problem chal-

lenging. However, GridFM-v0 can be hierarchically integrated into optimization schemes where generator power is iteratively adjusted, solution feasibility is checked, and other variables are determined using the power flow solver.

Long-term vision: GridFMs for a broader set of applications in power systems

In the “near-term: GridFM-v0, a GridFM for power flow applications” section, we explored an initial version of GridFM for power flow applications, already addressing important challenges of the industry. The long-term objective, however, is to address as many applications from Figure 4 as possible, using a single GridFM or a family thereof, fully exploiting FM capabilities. Here, we discuss how to build new industry-relevant applications on top of GridFM-v0.

Extending GridFM-v0 to a GridFM capable of handling new applications

Forecasting. GridFM-v0 was pre-trained to reconstruct power flow data for a single time step. This can be readily extended to pre-training to reconstruct power data over multiple consecutive time steps, similar to Weather FMs.⁷³ This enhancement will enable the forecasting of future grid states. By incorporating weather and climate data into the node embeddings, we will further improve the model’s ability to predict generation and anticipate weather-induced grid disturbances.

Transient and dynamic stability analysis. In addition to training GridFMs on static power flow data, such as in the case of GridFM-v0, the pre-training can additionally involve reconstructing high-resolution real-time data obtained from PMUs, which enables effective transient-state estimation. This approach will facilitate rapid fault current detection and localization, efficient system restoration, and disturbance mitigation.⁹⁸ Additionally, the generator rotor angle and voltage differential-algebraic equations can be added to the model using physics-based ML techniques⁹⁹ to, e.g., assess the grid’s stability margins.

Control operations. There are two concrete ways in which GridFMs will help in control operations. First, the generative

capabilities of FMs, such as serving as conditional load and generation scenario generators, can be used to construct scenario trees that are inputted to multistage stochastic programming algorithms, as done in Heitsch et al.¹⁰⁰ and Grove-Kuska et al.¹⁰¹ Second, we can use GridFM-v0 and GridFMs in general as digital twins of power grids within reinforcement learning (RL) environments. Leveraging digital twins (data-driven representations of systems) enables faster computation of state transitions during RL training. This also allows RL agents to be trained on systems with highly complex dynamics that are difficult to model explicitly, such as distribution networks. As demonstrated in Hua et al.,¹⁰² a digital twin of a distribution network was used to train an RL agent, improving decision-making in both planning and operations. Third, GridFMs can also be used as a pre-trained representation for RL agents, for which FMs have been shown to be a valuable asset.¹⁰³

Expansion planning. The pre-training of GridFM-v0, which involves reconstructing node-level masked power flow data, can be complemented by edge masking and reconstruction. Extending this approach further, we will leverage the model's generative capabilities to also propose new grid topologies that optimize for capacity expansion. By integrating constraints such as network reliability, load growth forecasts, and economic cost factors, GridFM can generate topologies that minimize congestion, maximize redundancy, and ensure efficient power distribution. Optimizing grid topology and resource dispatch co-optimization has already been achieved with GNNs in Authier et al.¹⁰⁴

Cybersecurity. GridFMs can be fine-tuned for anomaly detection tasks to identify patterns that signal potential cyber intrusions or physical failures.¹⁰⁵ We can also use GridFMs to identify vulnerabilities and conditions that lead to cascading failures.¹⁰⁶

Although the use cases mentioned above are the most apparent, experience with FMs and LLMs shows that once a model is available, numerous new applications can emerge. Therefore, the reader is encouraged to begin considering these possibilities because FMs for the power grid will soon become a reality.

Building multi-modal GridFMs

Ideally, GridFMs should be able to handle more than one set of application tasks. This can be done by combining multiple GridFMs together¹⁰³ or by building self-contained stand-alone larger models, ingesting multi-modal spatial, temporal, and text data.¹⁴ Finally, concepts such as mixture-of-experts (MoE) frameworks¹⁰⁷ can be exploited to compose GridFMs from a family of GridFMs.

Regardless of the method used, GridFMs will need to handle diverse data modalities in the long term, to account for complex external factors—such as climate, weather, economic and societal influences, distributed power sources, and the electrification of mobility and heating.⁷⁶ Training multi-modal FMs requires creating tokens for each modality, aligning them in space and time, and potentially normalizing or reducing their dimensionality.¹⁰⁸ Consequently, tasks beyond power flow and grid topology reconstruction¹⁰⁹ will need to be incorporated for pre-training. This approach has been demonstrated with models such as 4M,¹¹⁰ which unifies different modalities into discrete tokens for computer vision. We anticipate that

the number of trainable parameters for such models could approach those of weather models, reaching hundreds of millions.^{56,111}

CONCLUSIONS AND DISCUSSION

The rapid emergence of FMs is reshaping the AI landscape. Although FMs are well established for text and image analysis, here, we have introduced the idea of GridFMs: FMs for the power grid, able to learn from grid-relevant data to improve power grid operations, planning, and control. We first highlighted the strengths of FMs, which include excellent token prediction skills (signal reconstruction and forecasting), broad adaptability and scalability, and simulation acceleration. To demonstrate that GridFMs are well suited to close the growing gaps between computational capabilities and the industry's need to cope with the increasing uncertainty and complexity of the power grid, we draw on the experiences of FMs for weather forecasting. The latter is a domain characterized by apparently intractable complexity. Nevertheless, we have seen scalable and adaptable FMs to produce much more computationally efficient models by more than 4 orders of magnitude than traditional weather forecasting models.

We have outlined concrete implementation phases to create an initial version, GridFM-v0, which is pre-trained on power flow data and tailored to accelerate power flow-related downstream applications (e.g., optimal power flow and contingency analysis). In the same way that FMs have been able to learn the fundamental equations governing weather, we envision GridFM-v0 to learn power flow, thereby creating a much more efficient model. The key advantage of such a GridFM-v0 is that once pre-trained on multiple grids and various networks and operating conditions, it can be readily fine-tuned to specific grid topologies and user requirements based on their own private data.

This paper advocates for research and development of FMs for the power grid to facilitate the energy transition through collaboration between the various stakeholders. All stakeholders will benefit from GridFMs simply by fine-tuning it to their specific needs and with their proprietary data and knowledge—without re-developing a full bespoke AI model each time. Toward this end, we envision, under the leadership of industry associations and open source initiatives, for the community of stakeholders to join forces and make the required investments to research and develop GridFMs in open source, thereby revolutionizing power grid analysis and leveraging AI advancements to meet common goals and needs.

ACKNOWLEDGMENTS

The work was supported by the respective organizations of the authors. The authors acknowledge the invaluable discussions and interactions in the working group for Foundation Models for the Power Grid, including Roya Ahmadi, Rich Argentieri, Gianni Vittorio Armani, Venkat Balachandran, Ranjini Bangalore, Venkat Banunarayanan, Edmon Begoli, Larry Bekkedahl, Keith Benes, Michelle Berti, Budhu Bhaduri, Roberto Bianchessi, Gilbert Bindewald, Louise Blais, David Bromberg, Matthias Bucher, Eren Çam, Supriya Chinthavali, Julien Chosson, Joseph Collum, Magnus Cormack, Peter De Bock, Fei Ding, Kathy Duviella, Darlain Edeme, Yamshid Farhat, Mark Fenton, Marcus Freitag,

Alirez Ghassemian, Tassos Golnas, Gavin Goodland, Luca Grella, John Grosh, Benois Grossin, Andy Hackett, Hagen Haentsch, Zeryai Hagos, Grahame Hay, Frank Hellmann, Andy Henton, Raffael Hilber, Dan Hilton, Gabriela Hug, Alex Hurley, Sorana Ionescu, Kevin Johnston, Gareth Jones, Adrian Kelly, Bernd Klockl, Teddy Kott, Benjamin Kroposki, Teja Kuruganti, Larry Light, Dalton Lunga, Matt Magill, Adrian Maldonado, Antoine Maret, Pete Marsh, Russel Mason, James McClean, John Messer, Nikki Militello, Jordan Murkin, Xabi Muruaga, Indu Nambiar, Christian Nauck, Caleb Northrop, Gabby Nyirjesy, Marco Padula, Patrick Panciatici, Raul Alfaro Pelico, Mark Petri, Pierre Pinson, Abhishek Pontis, Josh Power, Alan Pressman, Feng Qiu, Mark Rafferty, Pippa Robertson, Nicoletta Rocca, Caroline Rose-Newport, Lyndon Ruff, Bryan Sacks, Kelly Sanders, Giovanni Sansavini, Michael Sheppard, Robert Shorten, Vikas Singhvi, Pratik Sonthalia, Brian Spears, Declan Stock, Olle Sundstroem, Matthew Tardugno, Roger Thompson, Anna Carolina Tortora, Alex Vivas, Lucy Yu, Ben Weiner, Casey Werth, Daniel Wetzl, Gary White, Xin Wu, Steven Wysmuller, Yunchi Yang, Thomas Zadlo, Marzia Zafar, and many others. This material is based upon work supported by the US Department of Energy, Office of Science, under contract number DE-AC02-06CH11357.

AUTHOR CONTRIBUTIONS

Conceptualization, H.F.H., B.G., L.S.A.M., A.P., A.V., and J.W.; investigation, H.F.H., B.G., L.S.A.M., A.P., A.V., J.W., J.B.-M., A.B.M., T.B., S.L.C., I.F., B.-M.H., R.J., K.K., V.M., F.M., M.D.M., O.R.-L., H.S., L.X., E.-N.S.Y., and A.Z.; project administration, H.F.H.; writing—original draft, H.F.H., B.G., L.S.A.M., A.P., A.V., J.W., J.B.-M., A.B.M., T.B., S.L.C., I.F., B.H., R.J., K.K., V.M., F.M., M.D.M., O.R.-L., H.S., L.X., E.-N.S.Y., and A.Z.; writing—review & editing, A.B., R.J.B., B.P.B., J.S., and S.S.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work, the authors used ChatGPT to find more concise reformulations and to condense paragraphs. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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